

Yufeng (Alex) Xia

xyf2002.github.io linkedin.com/in/yufeng-xia-12648a25b
Edinburgh, UK

Email: yufeng.xia@ed.ac.uk

Mobile: (+44) 7918155093

EDUCATION

- **University of Edinburgh: MSc by Research, Computer Science** Edinburgh, UK
ICSA; supervised by Prof. Mahesh K. Marina Sep 2025 – Nov 2026 (expected)
 - **Thesis:** *Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN.*
- **University of Edinburgh: BSc Computer Science** Edinburgh, UK
Grade: First Class Honours Sep 2021 – May 2025
 - **Relevant coursework:** Machine Learning Systems (A), Computer Architecture and Design (A), Computer Systems (A).

PUBLICATIONS

- EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** MobiCom'26 (cond. accepted)
Ujjwal Pawar, Andrew E. Ferguson, Yufeng Xia, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic
- Morphling: Emulator for Distributed Machine Learning at the Edge** EdgeSys'26 (Best Paper)
Leyang Xue, Yufeng Xia, Ismaeel Bashir, Eren Mendi, Jiaxun Yang, Myungjin Lee, Mahesh Marina
- Wasp: Foundation Model Training System for the Edge** SEC'26
Leyang Xue, Yufeng Xia, Myungjin Lee, Mahesh Marina
- CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization** under submission
Yufeng Xia, Pei Mu, Antonio Barbalace
- Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** under submission
Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, Yufeng Xia, Mun Choon Chan, Myungjin Lee, Mahesh Marina

RESEARCH EXPERIENCE

Compilers and Parallel Execution

- **CoSenseAQ: Automatic Quantisation in LLVM from Sensor Specifications** under submission
Junior Research Assistant, ICSA; first author; supervised by Dr Antonio Barbalace Apr 2024 – May 2025
 - **Transformation derived from a specification, not from source:** Built an LLVM framework that derives a quantisation strategy from sensor technical specifications, information a compiler cannot recover from the program itself, and applies it as IR-level transformations, replacing manual per-target tuning. Implemented as LLVM passes in C++ over a cross-compilation toolchain.
 - **Correctness and measured cost:** Established that the transformation stays within the numerical behaviour the specification permits; measured on ARM Cortex-M0 and Cortex-A53 in code size, execution time, power and area.

Systems for Machine Learning

- **Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN** MSc thesis
University of Edinburgh, sole author Sep 2025 – Present
 - **Partitioning and shedding as one decision:** Compute-sharing designs assume the harvesting workload is stateless and disposable; this one is neither, so reclaiming its compute cannot mean killing it. Formulated work placement as a joint objective across mutually interfering units and designed a greedy heuristic deciding which work is worth doing; the same ranking fixes the shedding order when compute is reclaimed mid-solve. A control loop answers resource pressure with graded actions (relocation, fidelity reduction, suspend and resume) rather than a single eviction.
- **Morphling: Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026, Best Paper
Research Assistant: C++ backend, fault tolerance, GPU partitioning Oct 2025 – Mar 2026
 - **Per-GEMM SM partitioning:** Implemented trace-driven GPU capacity partitioning with CUDA green contexts, dispatched through autograd hooks at per-GEMM granularity, so co-located training workloads share a device under explicit rather than incidental allocation.
 - **Event-driven backend and failure handling:** Built the proxy server, client and device-tracking layer in C++ on an epoll event loop with Protobuf messaging, supporting up to 1,800 emulated devices on one GPU server; added failure detection and target reselection, partition re-derivation, and runtime membership changes, removing a class of failure-induced deadlocks.
- **Weaver: Foundation Model Training over Shared AI Compute** under submission
Research Assistant: controller evaluation, profiling, large-scale runs 2024 – 2026

- **Harvesting accelerator capacity without disturbing a co-tenant:** Evaluated the controller deciding how much accelerator capacity a training workload may harvest from infrastructure shared with a latency-critical tenant. On a multi-site testbed it harvests up to **83%** of the available spare compute with no degradation to the co-tenant; characterised their interference through GPU profiling and ran the large-scale distributed training used in the evaluation.
- **EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation** MobiCom 2026, cond. accepted
Research Assistant: owned the end-to-end evaluation 2024 – 2026
 - **Kernel-level virtualisation, evaluated at scale:** Built and deployed a custom Linux kernel module and a modified KVM module providing time-dilated virtual machines, with automated provisioning, mesh routing and Kubernetes formation; ran the system across **130 cloud instances and 6,400 CPU cores**, emulating over **10,000** devices, and designed the fidelity and scalability evaluation.

TECHNICAL SKILLS

- **Languages** C, C++ (primary), Python, Verilog, Bash, Java.
- **Compilers & Kernels** LLVM pass development, IR-level analysis and transformation, automatic quantisation, cross-compilation toolchains; Triton kernel development (tiling, masking, memory coalescing), CUDA green contexts and SM partitioning, GPU profiling.
- **Systems** Distributed runtimes in C++ (epoll event loops, Protobuf messaging), scheduling under contention and preemption, Linux kernel modules and custom kernel builds, KVM virtualisation, multi-node cloud deployment (Azure, GCP, CloudLab); Docker, Kubernetes, Git.
- **Data** SQL, relational schema and index design, cache–database consistency in production (MySQL, Redis).
- **Architecture & ML** RTL design in Verilog, FPGA synthesis and on-board validation (Vivado, PYNQ-Z2), quantitative evaluation of CPI, cache behaviour and memory bandwidth; PyTorch, distributed foundation-model training, evaluation design.

AWARDS

- **Best Paper** EdgeSys 2026 (co-located with ACM MobiSys 2026), for *Morphling*.
- **The Edinburgh Award** University of Edinburgh, in recognition of professional development as a research assistant.

ACADEMIC SERVICE

- **EuroSys 2026** Student volunteer, on-site conference operations.

INDUSTRY EXPERIENCE

- **Senior Tech, Suzhou, China** Back-end Development Intern (Jun–Sep 2023): production backend APIs for a live delivery platform in a commercial engineering team: relational schema and index design, the SQL behind the business logic, and a Redis cache kept consistent with the database of record; platform migration to Spring Boot, code review and release testing (Java, MySQL, Redis, Kafka).

LANGUAGES

- **Mandarin** Native. **English** Fluent (IELTS 7.0, GRE 336).

REFERENCES

- **Prof. Mahesh K. Marina** MSc by Research supervisor, School of Informatics, University of Edinburgh • mahesh@ed.ac.uk
- **Dr Antonio Barbalace** BSc dissertation supervisor and Junior Research Assistant supervisor, School of Informatics, University of Edinburgh • antonio.barbalace@ed.ac.uk